

Java Programming - Basic Concepts

Java is a high-level, object-oriented programming language. It is widely used to build desktop applications, web applications, mobile applications, and enterprise software.

Inheritance

Java supports inheritance. The `extends` keyword is used to create an inheritance relationship between classes. In inheritance, a child class can acquire properties and methods from a parent class.

Encapsulation

Encapsulation is the process of wrapping data and methods together inside a class. The `private` keyword can be used to restrict direct access to variables from outside the class.

Polymorphism

Polymorphism allows an object to take multiple forms. It is commonly implemented through method overloading and method overriding.

Method Overriding

Method overriding occurs when a subclass provides its own implementation of a method inherited from its parent class. The `@Override` annotation is commonly used to indicate that a method is being overridden.

Abstraction

Abstraction focuses on hiding implementation details and showing only the essential functionality to the user. In Java, abstraction can be implemented using abstract classes and interfaces.

Important Java Keywords

`extends` - used for class inheritance.

`private` - restricts direct access to a class member.

`implements` - used when a class implements an interface.

`this` - refers to the current object.

`super` - refers to the parent class.